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Losartan shows limited benefit in preclinical models of Geleophysic dysplasia

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Abstract

Geleophysic dysplasia (GD) is a rare genetic disorder characterized by short stature, joint contractures, and cardiopulmonary complications, with early mortality, and linked to mutations in *ADAMTSL2* (GD1), *FBN1* (GD2), or *LTBP3* (GD3) genes. These mutations are hypothesized to disrupt extracellular matrix (ECM) organization and enhance transforming growth factor beta (TGF-β) signaling. Losartan, an angiotensin II receptor blocker, has been proposed to mitigate TGF-β-mediated pathologies. In this study we tested the efficacy of losartan as a therapeutic drug for GD. We evaluated losartan's therapeutic potential using *Adamtsl2* p.A165T mutant mice and patient-derived fibroblasts. Survival, growth, TGF-β signaling, and ECM protein expression were assessed. Losartan did not improve survival or growth in our mutant mice. Compared with control fibroblasts, patient-derived fibroblasts showed reduced basal TGF-β1 secretion. Consistent with this finding, transcriptomic analyses did not reveal activation of the TGF-β signaling pathway, and no differences in SMAD phosphorylation were observed between patient and control cells. Losartan treatment failed to modulate TGF-β signaling or ECM protein incorporation. These results suggest limited benefits of losartan in GD and challenge the notion of TGF-β dysregulation in GD pathogenesis, indicating a need for alternative targeted therapies.

Introduction

Geleophysic dysplasia (GD) is a progressive genetic disorder presenting with short stature, short fingers and toes, joint contractures, distinctive facial features, and cardiopulmonary complications that contribute to poor prognosis. [1-7] To date, three genes have been linked to GD: Geleophysic dysplasia type 1 (GD1, GPHYSD1, OMIM 231050) is an autosomal recessive form caused by mutations in *ADAMTSL2* (OMIM 612277). [8] GD type 2 (GD2, GPHYSD2, OMIM 614185) is autosomal dominantly inherited resulting from mutations in exons 41 or 42 of *FBN1* (OMIM 134797). [9] GD type 3 (GD3, GPHYSD3, OMIM 614185) is also autosomal dominant and caused by variants in *LTBP3* (OMIM 602090). [10]

ADAMTSL2 encodes a secreted matricellular glycoprotein within the ADAMTS (A Disintegrin And Metalloprotease with Thrombospondin motifs) superfamily, but it lacks the metalloprotease domain typical of this group. [11-13] Although the function of ADAMTSL2 is not fully understood, emerging evidence points to its role in organizing fibrillin microfibrils [14-17] and modulating the availability of Transforming Growth Factor Beta (TGF-β) via interaction with extracellular matrix (ECM) proteins such as FBN1 and LTBP1. [8,9,18-21] Fibrillin-1, encoded by the *FBN1* gene, is a large, calcium-binding ECM glycoprotein and a key structural component of microfibrils, with a regulatory role in the bioavailability of molecules like TGF-β. [20-24] *FBN1* mutations can cause Marfan syndrome, [25,26] but variants, in exons 41-42, which encode the TGF-β-binding protein-like domain 5 (TB5), can result in GD2. [8,9,26] These mutations are thought to contribute to GD2 through disruption of TGF-β signaling. Latent Transforming Growth Factor Beta Binding Protein 3 (LTBP3) is an extracellular matrix protein that regulates the availability of TGF-β. [27,28] Variants in *LTBP3* have been linked to genetic disorders with skeletal dysplasia, including GD3. [10,29,30]

Four major mechanisms have been proposed in GD pathogenesis: First, mutations in *ADAMTSL2*, *FBN1* and *LTBP3* genes have been shown to impair protein secretion disrupting the ECM organization. [8,28,31-33] Second, an increased TGF-β activity has been observed in cells from individuals with GD. [8-10,15,16,18,34,35] Third, our recent findings implicated matrix metalloproteinases (MMPs), which are a family of enzymes responsible for ECM remodeling, as potential contributors to GD pathology. [36] And four, a reduction in the Wnt signaling pathway was reported using a global gene knockout model. [37]

A clear genotype-phenotype correlation in patients with GD has not yet been established. However, our recent work demonstrates that the clinical severity of GD1 correlates with the abundance of ADAMTSL2 in the extracellular matrix. [31] Using both cellular and murine models that replicate the genetic profile of a GD patient compound heterozygous for two *ADAMTSL2* variants, *p.R61H* and *p.A165T*, we observed the impaired secretion of ADAMTSL2 in patient-derived dermal fibroblasts and in HEK-293T overexpression systems.[31] Notably, the p.A165T variant resulted in a more pronounced secretion defect.[31] Mice homozygous or hemizygous for the p.A165T variant exhibited growth impairment, respiratory and cardiac dysfunction, and early mortality.[31]

Losartan, an angiotensin II type 1 receptor blocker, has been proposed to ameliorate Marfan syndrome pathology through inhibition of TGF-β signaling. [38] Losartan could hypothetically offer therapeutic benefit in GD by modulating TGF-β pathway. In this study, we evaluated the effects of losartan in both *in-vivo* and *in-vitro* models of GD1, using optimized dosing strategies in a mouse model carrying the p.A165T *ADAMTSL2* variant, as well as in patient-derived fibroblasts. Our goal was to determine whether losartan could improve survival or growth in the animal model or restore the ECM organization in patient cells.

Results:

Losartan treatment does not improve the survival of Adamtsl2 mutant mice

To test the effects of losartan *in-vivo*, we utilized our previously generated GD animal models. Mice that were hemizygous for the *Adamtsl2* p.A165T missense variant were selected for treatment due to their phenotypic severity, especially their reduced survival and stunted growth. [31] Initially, we followed the treatment scheme and dosage previously used in mouse models of Marfan syndrome, consisting of prenatal and postnatal losartan treatments of 0.6 g/liter in the drinking water. [39] We found an increased lethality in the offspring of mice regardless of genotype with less than 40% survival in the first month (**Figure S1A**). This treatment scheme was not continued because of the high toxicity in WT mice.

The toxicity of losartan for the offspring during gestation is well documented; it can cross the placenta barrier, and it is secreted with the milk. For these reasons, losartan is a category D pregnancy risk medication and contraindicated in pregnant women.[40-44] Such toxicity was not found in previous studies with the Marfan syndrome mouse models; a possible explanation could be variations in the water intake among mice strains. [45,46] An adult mouse depending on the strain can drink from 2.5 to 50 mL of water per day. [45,46] Adult C57BL6 female mice with 25 g of weight, the same strain of our GD mice, drink around 6.4 mL water per day. [45] Most of the losartan toxicity studies in animals have been done on rats. The maximum dose in rats was calculated to be 10 to 20 mg/kg/day, and in humans 1 to 3 mg/kg/day. [40,47] In mice, previous studies have used losartan within 10 to 42.5 mg/kg/day. [40,48,49] Based on this information, we calculated that a dose 0.6 g/L in the drinking water provides 153 mg/kg/day of losartan in our animals, which is well above the estimated maximum dose. We proceeded to do a dose curve with our WT animals to determine the dose of losartan (from E18 to postnatal day 20) that could be tolerated without a significant effect in survival (**Figure S1B**) and found that 0.15 g/L in their drinking water provides a 38.25 mg/kg/day of losartan and has a minimal toxicity (**Figure S1B**). Therefore, we used this optimized dose to explore the potential therapeutic effect of losartan in the GD animal models.

Our previous data show that p.A165T hemizygous mice (A165T/-) present with a significant reduction in survival (~40%) after 3 months. [31] Most of the lethality was observed in the first few days after birth. When we treated mice with the losartan dose of 38.5 mg/kg/day from E18, we did not find a significant improvement in survival in the p.A165T hemizygous offspring (**Figure 1A**). Weight was monitored on days 5.5, 15.5, 22.5, and 36.5. Hemizygous p.A165T animals showed significantly reduced weight, starting from 15.5 days of age in males and females, when compared to WT, regardless of the treatment status (p<0.001, p<0.001, and p<0.001 in treated p.A165T hemizygous males at 15.5, 22.5, and 36.5 days, p=0.07, p<0.001, and p=0.002 in treated females p.A165T hemizygous at 15.5, 22.5, and 36.5 days) (**Figure 1B**). These results indicate that using losartan in this treatment modality does not have a beneficial effect on the survival or growth of GD1 mouse models.

Fibroblasts from patients with GD1 and GD2

Previous studies, including ours, have shown that mutations in the *ADAMTSL2* and *FBN1* genes impair the secretion of these proteins. [8,9,12,31] GD006 patient fibroblasts (GD1-2) were kindly provided by the Bristol Genetics Laboratory at North Bristol NHS Trust, Severn Pathology, Southmead Hospital, UK. To further investigate the cellular and molecular consequences of these variants, we analyzed primary human dermal fibroblasts obtained from patients with *ADAMTSL2* or *FBN1* mutations. Fibroblasts from healthy individuals were used as controls. Individuals with *LTBP3*-related GD (GD3) were excluded from this study due to the rarity of this subtype, which accounts for less than 1% of the GD cases. [1] All fibroblasts were isolated using the explant technique, as previously described. [31,36] The gene variants identified in the fibroblast samples are summarized in **Table 1** and were confirmed by sequencing of the *ADAMTSL2* and *FBN1* genes. GD1 patients carried compound heterozygous mutations in the *ADAMTSL2* gene, while GD2 patients had heterozygous mutations in the *FBN1* gene.

Age-matched fibroblasts were difficult to obtain. We used GSE113957, a high-coverage RNA-seq dataset from the NCBI Gene Expression Omnibus (GEO) featuring skin fibroblast cell lines from 133 healthy individuals (aged 1–94) and analyzed the RNA expression for the four main ECM proteins included in this study: ADAMTSL2, FBN1, fibronectin and Col1A1. Gene expression was stable across age as shown in **Figure S2**. (https://www.ncbi.nlm.nih.gov/geo/geo2r/?acc=GSE113957)

GD fibroblasts do not show activation of TGF-β pathway

To assess TGF-β production, we used an ELISA assay under three different culture conditions using DMEM medium as the base: 10% FBS, 1% FBS, and serum-free (0% FBS). Although all conditions yielded similar trends, we chose the serum-free condition for subsequent analyses to eliminate potential cross-reactivity from bovine TGF-β and to evaluate TGF-β1 secretion without external stimulation. On average, the active form of TGF-β1 comprised of only about 1% of total TGF-β1 secreted from the cells (**Figure 2A**). The control cells secreted approximately 200 pg/mL over three days. In comparison, patient cells secreted an average of 114 pg/mL, indicating that fibroblasts from patients secreted significantly less amount of total TGF-β1 than controls.

We next examined downstream TGF-β signaling by assessing SMAD2 and SMAD3 phosphorylation, under serum free conditions. Consistent with the low levels of active TGF-β, phosphorylated SMAD2 (S465/467) and SMAD3 (S423/425) were barely detected in control and patient's fibroblasts (**Figure 2B, 2C,** and **Figure S3)**. Total SMAD2 was at similar levels between the cells, while patient fibroblasts showed slightly reduced levels of total SMAD3 and SMAD4 (although not significant) (**Figure 2B, 2C**). Phosphorylated SMAD1/5/8 (S463/465) was undetectable in all samples, and total SMAD1 levels did not differ between the two groups (**Figure 2B, 2C**).

To get a deeper insight into TGF-β pathway on patient cells, we analyzed the fibroblast RNA-Seq data to profile the cell transcriptome and to identify active genes and pathways. We identified all genes associated with the TGF-β signaling pathway based on the Gene Set Enrichment Analysis database (GSEA, see methods). The transcript expression for all these selected genes was analyzed from the RNA-Seq data from control and patient-derived fibroblasts. The gene clustering did not reveal any significant dysregulation of the TGF-β pathway members between groups. Among the 56 genes analyzed, none showed a ≥2-fold change in expression between patient cells and controls (**Figure S4**). Detailed transcript expressions for all genes are shown in **Table S1**. In a similar way, we next examined genes previously reported to be upregulated following TGF-β1 stimulation (GSEA) and the RNA-Seq data for our cells was analyzed. The gene clustering did not indicate a consistent upregulation of these genes in patient samples (**Figure S5**). Of the 116 genes evaluated, only one gene, *ZIC4*, was significantly upregulated in patient cells (**Table S2**). Conversely, three genes, *CFAP298*, *SLC8A1*, and *JADE1*, were significantly downregulated (≥2-fold), arguing against a generalized activation of TGF-β signaling (**Table S2**). Finally, we also analyzed genes reported to be downregulated in response to TGF-β stimulation (GSEA). No clear pattern of downregulation emerged from gene clustering and shown in **Figure S6**. Among the 176 genes assessed, 10 (*LAMA5*, *GATA6*, *TGM2*, *PLAT*, *SSX2IP*, *SYNE2*, *C4B*, *EZR*, *MET*, and *ALCAM*) were significantly downregulated in patient cells, while one gene, *MTUS1*, was significantly upregulated (**Table S3**). Collectively, these analyses do not support the hypothesis of TGF-β pathway activation in the patient-derived fibroblasts.

To evaluate whether patient fibroblasts retained the ability to respond to TGF-β, we treated both control and patient cells with exogenous TGF-β1 for 24 hours, then assessed SMAD2 phosphorylation at the S465/467 site, a key event in TGF-β signaling. As shown in **Figure 2D** and **Figure S7**, all patient cells responded robustly to TGF-β after 24 hours of stimulation, showing comparable levels of SMAD2 phosphorylation to controls. Similar results were observed after 24 hours (data not shown), indicating that TGF-β signaling pathway remains intact in patients` fibroblasts despite their reduced basal secretion of TGF-β1 or phosphorylation status of SMAD2 or SMAD3.

Stimulation of TGF-β signaling increases* ADAMTSL2, FBN1 *and* fibronectin *in fibroblasts

To investigate the role of the TGF-β signaling pathway in the regulation of GD-associated proteins, we treated control and patient-derived fibroblasts with TGF-β1 or BMP2 for seven days and then assessed the levels of ADAMTSL2, FBN1, and fibronectin. As the extracellular matrix is not well produced under serum-free conditions, we used the regular media containing FBS (10%) for all subsequent experiments.

To check the pathway activation, we measured SMAD2 phosphorylation in response to TGF-β1 treatment (**Figure 3** and **Figure S8**). No differences in p-SMAD2 or total SMAD2 were observed between untreated control cells and untreated GD1 patient fibroblasts, under these conditions (**Figure 3A, 3B**). TGF-β1 not only induced phosphorylation of SMAD2 but also the phosphorylation of Smad1/5/8, indicating the activation of both the canonical and non-canonical pathways, as previously reported. [50,51] In contrast, BMP2 selectively induced phosphorylation of Smad1/5/8, consistent with its known signaling pathway (**Figure 3**).

TGF-β1 treatment led to a marked upregulation of ADAMTSL2 expression in all patient fibroblasts and in one control line (although no significant), with increases observed both intracellularly and extracellularly (**Figure 3**). BMP2 also increased ADAMTSL2 expression, albeit to a lesser extent and only in a subset of the cells tested. Similarly, TGF-β1 increased FBN1 expression in both patient (although no significant) and control fibroblasts, while BMP2 elicited an insignificant increase only in control cells (**Figure 3**). Notably, the fold increase in FBN1 expression was greater in the patient fibroblasts, likely due to their lower baseline levels, as we previously described. [31,36] Fibronectin levels were elevated following TGF-β1 treatment in both patient and control fibroblasts, whereas BMP2 had no significant effect (**Figure 3**). All these results highlight the key role of TGF-β pathway in the ECM organization and suggest that uncontrolled inhibition of the pathway might lead to additional ECM dysfunction.

Losartan does not affect TGF-β signaling or ECM protein incorporation in patient fibroblasts

To investigate the effect of losartan on TGF-β1 signaling and/or the extracellular matrix, patient-derived fibroblasts were treated with varying concentrations of the drug. Although no evidence of TGF-β pathway dysregulation was observed in these cells, we examined the effects of losartan on the extracellular matrix organization and composition, as proposed by Piccolo et al. [34] This was motivated by the hypothesis that losartan might influence the extracellular matrix organization not only through TGF-β dependent mechanisms. We slightly increased losartan concentration to 300 μM compared to 200 μM, as previously described with human dermal fibroblasts. [34] We did not detect any detrimental effect on cell viability (data not shown). As shown in **Figure 4A**, losartan did not significantly alter total TGF-β1 levels in the conditioned media under 10% FBS conditions when compared to untreated controls. A slight increase in active TGF-β1 was observed at certain concentrations; however, these values remained below the detection limit of the ELISA and were therefore considered unreliable. At low concentrations, losartan did not affect the total TGF-β1 secretion in patient fibroblasts. Interestingly, treatment with losartan at 300 μM, resulted in a slight but significant increase in total TGF-β1 secretion in patient cells. Despite this increase, no differences in SMAD2 phosphorylation were observed upon losartan treatment, as demonstrated by western blot analysis (**Figure 4B** and **Figure S9**), suggesting that downstream TGF-β signaling remained unaltered.

Under similar conditions, we studied the effect of losartan on the extracellular matrix, which was isolated with a decellularization technique that removes cultured cells while preserving a clean, intact three-dimensional matrix attached to the culture surface (NH_4OH/Triton X-100 method). ECM derived from control fibroblasts displayed a dense, well-aligned, and highly organized fiber network, a structure that was not observed in the fibroblasts from the patient (**Figure S10**). [36] Furthermore, losartan treatment did not improve ECM organization in primary patient-derived fibroblasts, and a non-significant reduction (~20% at 300 μM losartan) in total fiber content was observed (**Figure S10**).

A more detailed analysis of the ECM proteins by western blot was explored, and we focused on proteins related to the GD pathology. Consistent with our previous findings, the ADAMTSL2 protein was predominantly detected in the intracellular (IC) compartment (bands-1 and band-2, **Figure 4B**) with limited detection in the ECM fraction (bands-3 and band-4), relative to IC levels. [31,36] GAPDH and fibronectin served as controls for IC and ECM fractions, respectively. *ADAMTSL2* mutations did not affect its intracellular expression, and no significant differences were observed between control and patient fibroblasts. However, secretion of ADAMTSL2 into the conditioned media and its incorporation into the ECM were significantly reduced in patient fibroblasts when compared with control cells. Treatment with losartan, even at increasing concentrations, did not alter ADAMTSL2 intracellular expression, secretion, or ECM incorporation (**Figures 4B, 4C**).

In contrast to ADAMTSL2, FBN1 was primarily localized to the ECM, with minimal intracellular localization. [31,36] Notably, the intracellular form of FBN1 exhibited a slightly higher molecular weight compared to its secreted form (**Figure 4B**). As previously reported, *ADAMTSL2* mutations led to reduced intracellular FBN1 levels in patient fibroblasts relative to controls (**Figures 4B, 4C**). Although secretion of FBN1 was modestly affected, differences were not statistically significant. The most pronounced alteration was a substantial reduction in ECM-incorporated FBN1 in patient cells. Losartan treatment did not modify FBN1 intracellular levels, secretion into the conditioned media, or incorporation into the ECM.

We further examined the expression and distribution of two additional ECM components, fibronectin and collagen type-I alpha 1 (Col1A1). Fibronectin was primarily detected in the ECM with limited intracellular presence, whereas Col1A1 was largely retained intracellularly with poor ECM incorporation. [31,36] In patient fibroblasts, fibronectin expression and secretion into the conditioned media were comparable to controls; however, ECM incorporation was markedly reduced. Col1A1 showed mildly increased intracellular expression in patient fibroblasts, but secretion levels were unchanged. Like fibronectin, Col1A1 incorporation into the ECM was slightly but not significantly impaired in the patient cells; suggesting that losartan concentration of 300 μM is sufficient to elicit biological effects on the cells. Importantly, losartan treatment had no observable effect on the intracellular expression or secretion of eighter analyzed protein (**Figure 4B, 4C**).

We verified the expression of angiotensin II receptor 1 (AT1) in the two controls (CT-1 and CT-2) and in the patient fibroblasts (GD1-1). As shown in **Figure 4D**, no difference was observed between the cells.

Discussion

Previous studies have suggested that dysregulation of TGF-β signaling is the underlying mechanism in GD. Fibroblasts derived from patients with mutations in *ADAMTSL2* and *FBN1* genes have consistently shown altered TGF-β signaling. [1,8,9,20,21,23,24] Although GD-associated mutations for the *LTBP3* gene similarly disrupt the microfibrillar network, it does not appear to increase TGF-β signaling in the dermal fibroblast model described for GD3.[1,10] Nevertheless, in a different cellular model for adipogenesis, the loss of LTBP3 resulted in a reduced adipogenesis via an increase in TGF beta signaling. [52]

In the present study, our *ADAMTSL2* p.A165T mouse model displayed severe growth impairment and high perinatal mortality, yet losartan treatment failed to improve either outcome. Unexpectedly, we observed no evidence of TGF-β pathway activation in dermal fibroblasts derived from GD patients. Compared to control fibroblasts, patient cells secreted lower levels of TGF-β1 and no difference in SMAD2/3 phosphorylation. Interestingly, in our previous study, transient transfection of ADAMTSL2 variants into HEK-293 cells did not modulate TGF-β signaling. [31] In this study, we included the two variants, p.R61H and p.A165T, with the latter significantly impairing protein secretion but showing no effect on TGF-β signaling. In addition, we reported no active TGF-β1 in the lung tissues of newborn *Adamtsl2-KO* animals compared to littermate WT; and there was no significant increase of active TGF-β1 in lung or heart tissues, in adult mutant animals, *Adamtsl2-R61H, Adamtsl2-A165T or Adamtsl2-R61H/A165T*, when compared to WT animals. [31] Supporting this, recent work in human cardiac fibroblasts demonstrated that overexpression of ADAMTSL2 did not inhibit TGF-β pathway activation when cells were treated with recombinant active TGF-β1. [18]

Treatment with active TGF-β induced phosphorylation of SMAD2 without significant differences between control and patient cells, suggesting that the signaling machinery remain intact. TGF-β1 and ADAMTSL2 are proposed to be involved in a bidirectional regulatory loop. [15,53] ADAMTSL2 inhibits TGF-β signaling by binding to LTBP1 within the microfibrillar network, thereby reducing the bioavailability of active TGF-β. [15,53] Conversely, TGF-β1 signaling can induce the expression of ADAMTSL2, suggesting a feedback mechanism wherein increased TGF-β activity promotes ADAMTSL2 expression to attenuate further signaling. [18,54] In this study, treatment with TGF-β1 and BMP2 led to increased expression of both ADAMTSL2 and FBN1 in all primary fibroblast. This suggests that upregulation of these ECM components by TGF-β1 is a generalizable mechanism in dermal fibroblasts, independent of the genotype.

Our findings suggest that an increase in TGF-β signaling might not be the primary driver of the disease pathology. In fact, there are evidences from other studies suggesting that an increased TGF-β signaling is not consistently observed across all models of GD. [15,31,37,55] For example, in *ADAMTSL2*-deficient myoblasts no increase in TGF-β activity was detected.[37] In *Adamtsl2* knockout mice, TGF-β signaling was not elevated at birth and pan-TGF-β neutralizing antibody failed to rescue the phenotype. [15]

A similar mechanism involving dysregulated TGF-β signaling has been proposed for Marfan syndrome, which is caused by *FBN1* mutations. [56-58] Marfan syndrome patients present a phenotype that is notably opposite to that of GD, characterized by tall stature and long extremities. This striking contrast raises the question of how mutations in the same gene and potentially overlapping molecular pathways can lead to such divergent clinical manifestations. Numerous studies have evaluated losartan's efficacy in Marfan syndrome animal models and patient cohorts,[59-65] yet no investigation has examined its effects in GD patients or animal models. Notably, losartan was reported in 2019 to improve lysosomal inclusions and microfibril deposition in GD2 primary fibroblast in vitro (*FBN1* mutation). [34] Motivated by this finding, and by recurrent inquiries from patients and clinicians about losartan as a potential therapy for GD, we investigated its efficacy in our GD models. Treatment with losartan failed to alter the levels of four key ECM proteins, including FBN1, ADAMTSL2, fibronectin, and Col1A1, suggesting a limited or a genotype-specific effect.

In our study, losartan conferred no therapeutic benefit in GD1 mice and failed to improve ECM organization in patient primary fibroblasts, suggesting that GD and Marfan syndrome do not share a common pathogenic mechanism.

Our findings might challenge the assumption that TGF-β dysregulation is a universal feature of GD and suggest that TGF-β-targeted therapies might not be effective in all GD cases. Further work is needed to elucidate the precise molecular pathways involved in GD and to identify more targeted therapeutic strategies.

Material and Methods

Study approval: All human and animal studies were reviewed and approved by the University of Miami Institutional Review Board (IRB) and by the University of Miami Institutional Animal Care and Use Committee (IACUC), respectively, and according to the National Institutes of Health guidelines (NIH). All patients were evaluated at the University of Miami or by collaborators. Written informed consents for the testing and sharing of the research data were obtained from all patients or their parents, including the use of the isolated primary fibroblasts. All animal studies and experiments were performed in accordance with relevant guidelines and regulations; and all reports were in accordance with ARRIVE guidelines.

Losartan potassium: Losartan was obtained from Tokyo Chemical Industry, Tokyo Japan. Diluted in the drinking water at 0.6, 0.3, 0.15, and 0.075 g/L to provide 153, 76.5, 38.25, and 19.125 mg/kg/day respectively.

***Adamtsl2* p.A165T knock-in mice:** For the generation of *Adamtsl2* p.A165T missense variant, we developed a knock-in mouse model carrying the variant by conventional embryonic stem cell-mediated knock-in technology using the service of Taconic-Cyagen Model generation Alliance (Germantown, NY), as previously described.[31] *Adamtsl2* p.A165T heterozygous mice were crossed between them to obtain homozygous animals or with heterozygous mice carrying a knock-out allele of *Adamtsl2* [14], to obtain hemizygous mice. Animals were genotyped as previously described and housed under a 12-hour light-dark cycle at 20-23 °C in specific pathogen-free facilities and supplied with food and water ad libitum. All studies and experiments were performed in accordance with relevant guidelines and regulations. In addition, all reports were in accordance with ARRIVE guidelines. All litters, excluding the first one, from different types of matings, were included in these studies until at least 10 animals per group was reached. Dams with high litter mortality caused by others factor like complications during birth were removed from the study. *Adamtsl2* p.A165T hemizygous mice were compared with *Adamtsl2* wild-type (WT).

Survival and growth curves: The survival up 20 days or up to 3 months was recorded. A Kaplan-Meier survival curve was done to visualize the survival rate during the first 3 months. To determine significant differences, the curves were analyzed with the Gehan-Breslow-Wilcoxon test that is better in determining significant difference in early time points. For growth curves, the weights at specific time points were recorded and plotted. Graph and student T-test were performed with Microsoft Excel (Redmond, WA). The number of animals is indicated in between parentheses in the figure.

Cell culture experiments: All human dermal fibroblasts were isolated from skin biopsies, following the explant technique, as previously described. [31,36] Briefly, tissues were cut into small explants around 1x1 mm, and incubated at 37°C and 5% CO_2 atmosphere, using complete Dulbecco's Modified Eagle Medium (DMEM) high glucose, pyruvate (Thermo Fisher Scientific, 11995-065), 20% FBS (Gen Clone, Genesee Scientific, 25-514), 100 U/mL of penicillin, 100 μg/mL of streptomycin (Corning, 30-002), and 200 μg/mL of Primocin (InvivoGen, ant-pm-05). The medium was changed every day until cells migrated from the explant and grew as a monolayer, and then every other day until the cells were cryopreserved as passage P2. Thawed fibroblasts for experiments were cultured in T75 flasks and used in a range from P4-P10 passages. All experiments were completed using DMEM, 10% FBS, 100 U/mL of penicillin, and 100 μg/mL of streptomycin.

Cells treatments: All cells were treated with 20 ng/mL TGF-β1 (Peprotech, 100-21-10UG), for 24 hours to check TGF-β pathway: p-SMAD2 (S465/467), SMAD2 and GAPDH. Cell lysates were obtained and analyzed by western blots. All cells, except GD2-3, were also treated with 20 ng/mL

TGF-β1 and 20 ng/mL BMP2 (Sigma-Millipore, H4791-10UG) for 7 days, and media with the drug was changed every other day, to analyze their effect on the extracellular matrix proteins: ADAMTSL2, FBN1, fibronectin and GAPDH. Total lysates were obtained and analyzed by western blots. GD1-1 cells were treated with losartan at 3.3, 11, 33, 100 and 300 µM for 7 days; and media was changed every other day. Conditioned media were collected from third to sixth day (3 days) for analysis of TGF-β1 and extracellular protein secretion. Intracellular lysates and extracellular matrix lysates from losartan-treated cells and untreated control cells (GD1-1 and CT-1) were obtained for western blot analysis, always including ADAMTSL2, FBN1, fibronectin and GAPDH proteins.

TGF-β1 quantification: TGF-β1 was quantified from conditioned media of dermal fibroblasts as previously described. [8,31,36] Briefly, 0.5 million cells were cultured in 12-well plates for a total of 7 days: first using complete DMEM high glucose, pyruvate, 10% FBS, 100 U/mL of penicillin, and 100 µg/mL of streptomycin. Media were changed every 2 days. After confluency (4 days later), cells were washed twice with PBS, and DMEM without FBS were added. Conditioned media were collected 72 hours later for further analysis. Human TGF-β1 DuoSet ELISA (R&D Systems, DY240), was performed according to manufacturer instructions.

Protein lysates and samples: Protein lysates were prepared using four different approaches depending on the experiment goal.

1) ***Cell lysates*** (CL) for the TGF-β1 pathway proteins were prepared using 1x RIPA lysis buffer (Millipore-Sigma, R0278) containing 50 mM Tris-HCl, pH 8.0, with 150 mM sodium chloride, 1.0% Igepal CA-630 (NP-40), 0.5% sodium deoxycholate, and 0.1% sodium dodecyl sulfate (SDS), plus a 1x protease and phosphatase inhibitor cocktail (Pierce Protease and Phosphatase Inhibitor Mini Tablets, EDTA-free, Thermo-Fisher Scientific, A32961). RIPA was added directly to the cells, to avoid the addition of fresh medium containing FBS and a possible activation of the TGF-β1 pathway. Then the cells were scraped and incubated on ice for 15 minutes. Cell lysates were cleared at maximum speed (14000 rpm) for 5 minutes.

2) ***Intracellular lysates*** (IC) were prepared by adding RIPA buffer to trypsinized cells and incubated on ice for 15 minutes. Trypsin-EDTA (0.25%, Gibco 25200056) was added to the cells and incubated at 37°C for 5 minutes. Trypsin was inhibited using complete medium DMEM (10% FBS) and cells were washed with PBS. Intracellular lysates were cleared at maximum speed (14000 rpm) for 5 minutes.

Protein concentration was determined using a Bradford colorimetric assay kit (Bio-Rad Protein Assay Kit-II Bio-Rad, 500-0002).

3) ***Total lysates*** (Total) were prepared by adding 2x Laemmli buffer (Bio-Rad, 1610747) containing 10% β-mercaptoethanol (Millipore-Sigma, M3148-100ML), directly to the cells in well, previously washed with PBS.

4) ***ECM protein lysates*** were prepared using the Ammonium hydroxide (NH_4OH) / Triton X-100 protocol. [66] Briefly, cells were washed with PBS and removed from the plate with 20 mM NH_4OH, 0.05% Triton-X-100 in PBS for 15 minutes at room temperature. The remaining ECM proteins attached to the plate (decellularized ECM) were washed with PBS and then pictures were taken with a 100x total magnification. Pictures of decellularized ECM were transformed into binary images (black and white) for visualization and quantification purposes using ImageJ. Representative images were selected from 5 independent experiments. The decellularized ECM was dissolved in 2x Laemmli buffer as 138.9 mM Tris-HCl, pH 6.8, 22.2% (v/v) glycerol, 2.2% LDS and 0,01% bromophenol blue containing 10% 2-mercaptoethanol for western blot analysis.

5) Conditioned media were collected using different medium composition and different time points as specified in experiments.

Western blot analysis: Ten to twenty µg of proteins were subjected to SDS-polyacrylamide gel electrophoresis (SDS-PAGE), using the appropriate gel, and transferred to 0.22 µm PVDF membrane (Bio-Rad, 1620112). After blocking with 5% BSA in Tris Buffered Saline (TBS-T20, 25mM Tris, 140 mM sodium chloride, and 3.0 mM potassium chloride, and containing 0.1% tween-20); the membranes were incubated with the corresponding primary antibody overnight at 4^{o}C (**Table S1**). Primary antibodies were detected by horseradish peroxidase conjugated secondary antibody and the immunocomplexes were visualized by chemiluminescence (Super Signal™ West, Thermo-Fisher Scientific, 34577 and 34096). The membranes were stripped using Restore™ PLUS Western Blot Stripping Buffer (Thermo Scientific, 46430) at room temperature, for 15 minutes, extensively washed with TBS-T20, blocked with 5% BSA in TBS-T20, and re-probed with the corresponding antibodies following a similar protocol.

Whole transcriptome sequencing (RNA-Seq): Total RNA was extracted from primary human fibroblast, at P4. The cells were grown in DMEM 10% FBS, but the serum was removed overnight the day before the collection. The RNA was extracted using the RNeasy Mini Kit (Qiagen). A Bioanalyzer 2000 was used to measure the quality of RNA. All samples' RNA integrity numbers (RIN) were above 9. Whole transcriptome sequencing (RNA-seq) was conducted at the Sequencing Core of John P. Hussman Institute of Human Genomics at the University of Miami using the TruSeq Stranded Total RNA Library Prep Kit from Illumina (San Diego, CA). Briefly, after ribosomal RNA (rRNA) was depleted, sequencing libraries were ligated with standard Illumina adaptors and subsequently sequenced on a Hiseq2000 sequencing system (100 bp single end reads, Illumina, San Diego, CA, USA). Trimming was performed, and sequence reads were aligned to the human transcriptome (GRCh38/hg38 assembly from the Genome Reference Consortium) and quantified using the STAR aligner. [67] All samples had uniquely aligned reads between 35,607,220 and 51,912,532. Read data was run through quality control metrics using MultiQC. [68] Statistical significance was first examined with three alternative differential expression calculators: edgeR, DESeq2, and Bayseq. [69-71] For visualization and further analysis iDEP software was used. [72] The program calculates CPM values from raw counts. To reduce false positive a minimum of 5 CPM per library was used. Data was transformed, normalized for clustering with rlog (transformation implemented in DESeq2). Of 60,656 genes from the data set, 11,147 genes passed the filter. A supervised Hierarchical and K-Means clustering with heatmap was generated to have a specific view of the expression of metalloprotease family of genes.

Gene Set Enrichment Analysis (GSEA): Database was used to identify all genes associated with the TGF-β signaling pathway, as well as genes associated with TGF-β1 stimulation, including reported up-regulated and down-regulated genes. [73,74]

Statistics: Data are expressed as mean ± SEM (standard error of the mean) and all graphs were performed in Microsoft Excel. The number of replicates for experiments are indicated in each figure and statistical differences were tested using the 1-way or 2-way Anova test when comparing samples or a group with the corresponding control(s). p-values < 0.05 were considered statistically significant as * $p<0.05$, ** $p<0.01$, *** $p<0.001$ and **** $p<0.0001$.

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Author contributions

AAM designed and conducted all cells experiments, acquired and analyzed data, and wrote the manuscript. VC designed and conducted animal experiments, acquired and analyzed data, and wrote and reviewed the manuscript. LP and SS recruited patients and reviews the manuscript. KW designed animal experiments analyzed data and reviewed the manuscript. GW analyzed data and reviewed the manuscript. MT designed the research study, analyzed data, reviewed and approved the manuscript.

Declaration of interests

The authors or any immediate family members have no relevant financial or non-financial interests to disclose. All author affiliations have been declared. All funding sources for this study are listed in the acknowledgments section of the manuscript. The authors and our immediate family members have no related patent applications or registrations to declare.

Data availability

The authors declare that the data supporting the findings of this study are available within the paper and its Supplementary Information files. Should any raw data files be needed in another format they are available from the corresponding author upon reasonable request. The datasets generated and/or analyzed during the current study are available in the ClinVar repository. Submission IDs for reported gene variants include: *ADAMTSL2* c.182G> A as SUB15022370, *ADAMTSL2* c.493G> A as SUB15022407, *ADAMTSL2* c.542 T> C as SUB15022419, *ADAMTSL2* c.707 C> T as SUB15022429 and *FBN1* c.5284G> A as SUB15022437, *FBN1* c.5183 C> T as SUB15022369 and *FBN1* c.5243G> C as SUB15022471. RNASeq data are deposited at GEO with accession number GSE292600.

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Figure Legends:

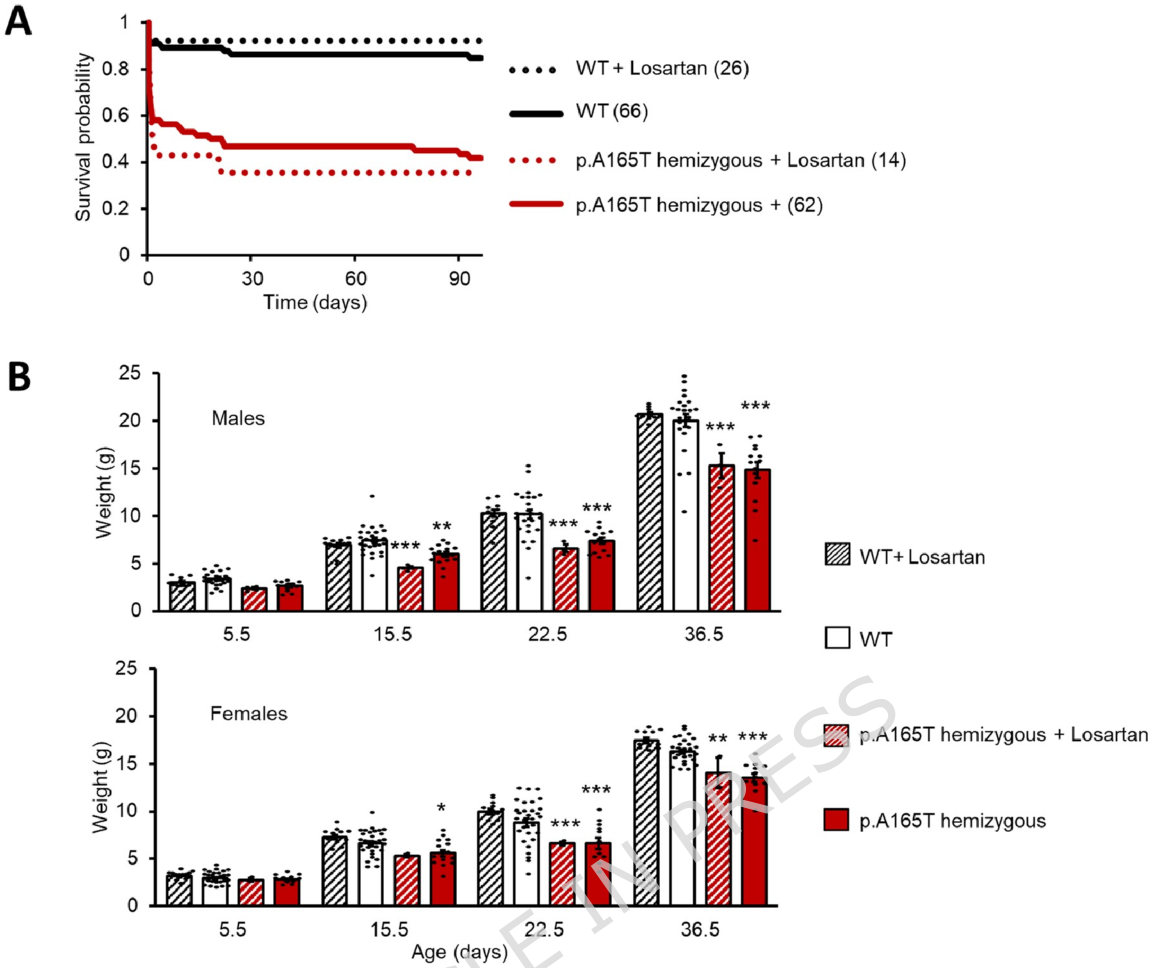
Figure 1- Losartan potassium does not improve survival or growth in a GD1 animal model. (A) Losartan treatment 38.25 mg/kg/day does not improve survival in *Adamtsl2* mutant mice. Kaplan-Meier survival curve of mice with p.A165T allelic variant with and without Losartan treatment during the first 3 months. The curves were compared with Gehan-Breslow-Wilconox test to determine significance. A significant reduction in survival during the first 3 months was expected in *Adamtsl2* p.A165T hemizygous (p<0.0001) mice vs. WT mice. The survival did not improve after losartan treatment of *Adamtsl2* p.A165T hemizygous. The treatment was administered in the drinking water of the dams at P18 and continued for 3 months postnatally. N=number of animals, WT *Adamtsl2*=black line, and p.A165T hemizygous=red line. No treatment=continue line and Losartan=dotted line. **(B)** Weight for male (top) and female mice (bottom). Average +/- SEM (*, p<0.05, ** p<0.01, and *** p<0.001).

Figure 2- TGF-β1 pathway is not significantly dysregulated in patient fibroblasts. **(A)** TGF-β1 was determined by ELISA, in conditioned media after 3 days culture in DMEM without FBS. Continued and dashed lines in bars represent the active and the total TGF-β1, respectively. Averages and standard error of the mean (SEM) were calculated for 5 independent experiments. Significance was calculated using a 1-way Anova test and values were compared versus the average of CT samples for the five independent determinations. **(B)** Proteins related to TGF-β pathway were determined in protein lysates by western blot: p-SMAD2 (S465/467), p-SMAD3 (S423/425), Smad4 and p-SMAD1/5/8 (S465/467). Representative images were selected from seven and four independent experiments for p-SMAD2 and others, respectively. **(C)** Densitometry analysis was completed using ImageJ software, for p-SMAD2/SMAD2, SMAD2/GAPDH, p-SMAD3/SMAD3, SMAD3/GAPDH, SMAD4/GAPDH, p-SMAD1/5/8 / SMAD1 (no signal for experiments) and SMAD1/GAPDH ratios of western blots in (B). Averages and standard error of the mean (SEM) were calculated. Significance was calculated using a 1-way Anova test and values were compared versus the average of CT samples for independent determinations. Original uncropped blots are shown in **Figure S3**, with a rectangle indicating the cropped area. **(D)** TGF-β pathway showed normal response after cells were stimulated with TGF-β1 for 24 hours and p-SMAD2 and SMAD2 were determined by western blot. GAPDH was used as loading control for all western blot studies. Original uncropped blots are shown in **Figure S7**, with a rectangle indicating the cropped area. Significance was represented as: * p<0.05, ** p<0.01, *** p<0.001, and **** p<0.0001.

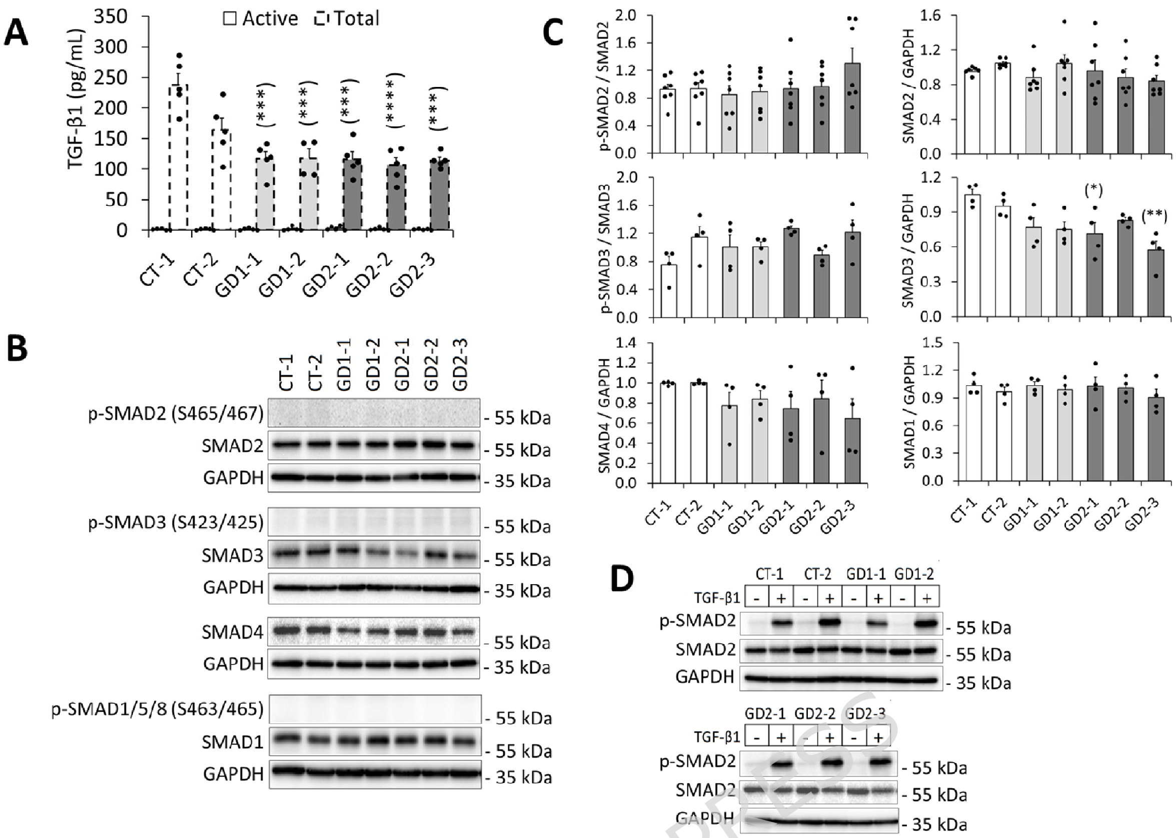
Figure 3- Exogeneous TGF-β1 strongly induces up-regulation of key proteins of the ECM: ADAMTSL2, FBN1 and fibronectin. **(A)** Control and patient fibroblasts were treated with TGF-β1 (20 ng/mL) or BMP2 (20 ng/mL) for 7 days. Proteins related to TGF-β and BMP2 pathways were

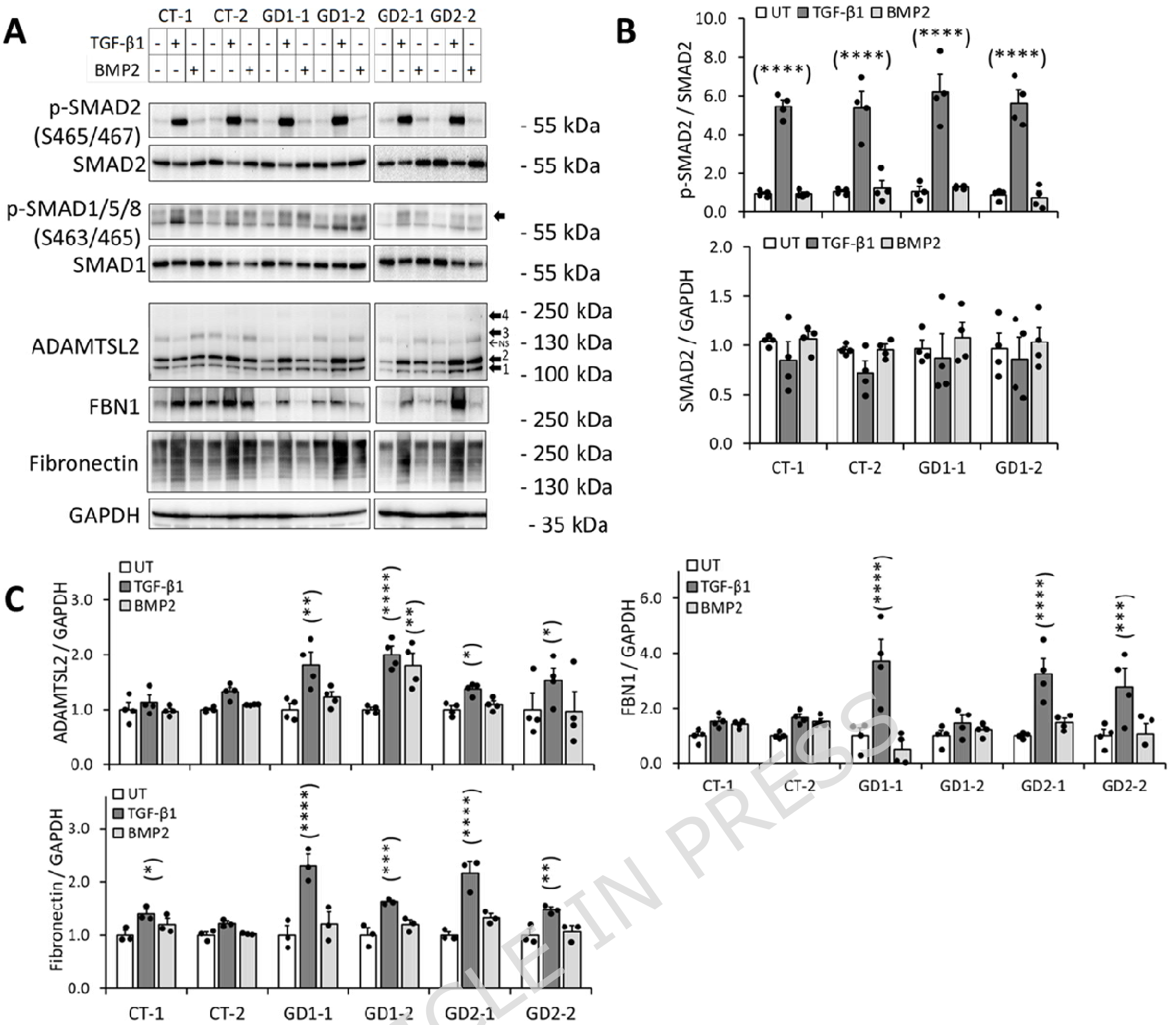
determined in Total lysates by western blot. Pathway activation was measured by p-SMAD2 (S465/467) and p-SMAD1/5/8 (S465/467), respectively; in addition to the ECM: ADAMTSL2, FBN1 and fibronectin. Representative images were selected from four independent experiments. Original uncropped blots are shown in **Figure S8**, with a rectangle indicating the cropped area. GAPDH was used as loading control. Individual cell fractions were used to identify the corresponding bands for ADAMTSL2 and FBN1 as represented in **Figure 4**, and as previously described. [31,36]. Band-1 and -2 correspond to the intracellular ADAMTSL2 forms, while band-3 and -4 correspond to the protein incorporated into the ECM. FBN1 was detected as a single band combining both intracellular and ECM forms. **(B)** Densitometry analysis for p-SMAD2 and total SMAD2 were completed using ImageJ software for the whole area covering all bands, and significance was calculated using a 2-way Anova test. Non-significant differences (NS) were found for p-SMAd2 or total SMAD2 when untreated control cells were compared vs untreated GD1 patient cells. Values for TGF-β1 (****) or BMP2-treated cells (NS) were compared versus the corresponding individual untreated (UT) value. **(C)** Densitometry analysis for the extracellular matrix proteins: ADAMTSL2, FBN1 and fibronectin were completed in a similar way. Values for TGF-β1 or BMP2-treated cells were compared versus the corresponding individual untreated (UT) value; and then all values were normalized by the individual sample average. As Total lysates were run on different membranes, densitometry analysis does not compare values between different samples. Significance was represented as: * $p<0.05$, ** $p<0.01$, *** $p<0.001$, and **** $p<0.0001$. Bars and error bars represent the average and SEM for all values, respectively.

Figure 4- Losartan does not affect TGF-β1 levels or ECM component in patient fibroblasts. Cells were treated with increasing concentration of losartan for 7 days. Cell lysates and ECM lysates were obtained after 7 days. **(A)** Conditioned media (CM) were collected from the third day to the sixth day (3 days culture) and TGF-β1 was determined by ELISA. Continued and dashed lines in bars represent the active and the total TGF-β1, respectively. Average and standard errors of the mean (SEM) were calculated for 5 independent experiments. No significance was obtained by a t-Student test when values were compared versus CT or untreated cells. **(B)** Intracellular lysates, ECM lysates as well as conditioned media (CM, 3 days) were used for western blot studies for ADAMTSL2, FBN1, fibronectin and Col1A1. GAPDH was used as loading control. Original uncropped blots are shown in **Figure S10**, with a rectangle indicating the cropped area. **(C)** Densitometry analysis was completed using ImageJ software, and all values were normalized by GAPDH. Averages and standard errors of the mean (SEM) were calculated for 4 independent experiments. Significance was calculated using a 2-way Anova test and values were compared first untreated patient (GD1-1) versus CT-1 fibroblasts; and second, all losartan-treated patient fibroblast (GD1-1) versus the untreated patient fibroblast (GD1-1). **(D)** Total lysates for CT-1, CT-2 and GD1-1 fibroblasts were used for western blot studies of the angiotensin II receptor-1 (AT1). GAPDH was used as a loading control. Densitometry analysis was completed using ImageJ software, and all values were normalized by GAPDH. Averages and standard errors of the mean (SEM) were calculated for 5 independent experiments (only 2 experiments for CT-2). Significance was represented as: * $p<0.05$, ** $p<0.01$, and *** $p<0.001$.



A
Survival probability
Time (days)
WT + Losartan (26)
WT (66)
p.A165T hemizygous + Losartan (14)
p.A165T hemizygous + (62)
B
Males
Females
Weight (g)
Age (days)
5.5
15.5
22.5
36.5
WT+ Losartan
WT
p.A165T hemizygous + Losartan
p.A165T hemizygous


A
TGF-β1 (pg/mL)
Active
Total
(***)
(****)
CT-1
CT-2
GD1-1
GD1-2
GD2-1
GD2-2
GD2-3
B
p-SMAD2 (S465/467)
SMAD2
GAPDH
p-SMAD3 (S423/425)
SMAD3
SMAD4
p-SMAD1/5/8 (S463/465)
SMAD1
- 55 kDa
- 35 kDa
C
p-SMAD2 / SMAD2
SMAD2 / GAPDH
p-SMAD3 / SMAD3
SMAD3 / GAPDH
(*)
(**)
SMAD4 / GAPDH
SMAD1 / GAPDH
D
TGF-β1
p-SMAD2
SMAD2
GAPDH


A
CT-1 CT-2 GD1-1 GD1-2 GD2-1 GD2-2
TGF-β1
BMP2
p-SMAD2 (S465/467)
SMAD2
p-SMAD1/5/8 (S463/465)
SMAD1
ADAMTSL2
FBN1
Fibronectin
GAPDH
- 55 kDa
- 55 kDa
- 55 kDa
- 55 kDa
- 250 kDa
- 130 kDa
- 100 kDa
- 250 kDa
- 250 kDa
- 130 kDa
- 35 kDa
4
3
NS
2
1
B
UT TGF-β1 BMP2
p-SMAD2 / SMAD2
(****)
SMAD2 / GAPDH
CT-1 CT-2 GD1-1 GD1-2
C
ADAMTSL2 / GAPDH
FBN1 / GAPDH
Fibronectin / GAPDH
(*) (**) (***) (****)
CT-1 CT-2 GD1-1 GD1-2 GD2-1 GD2-2

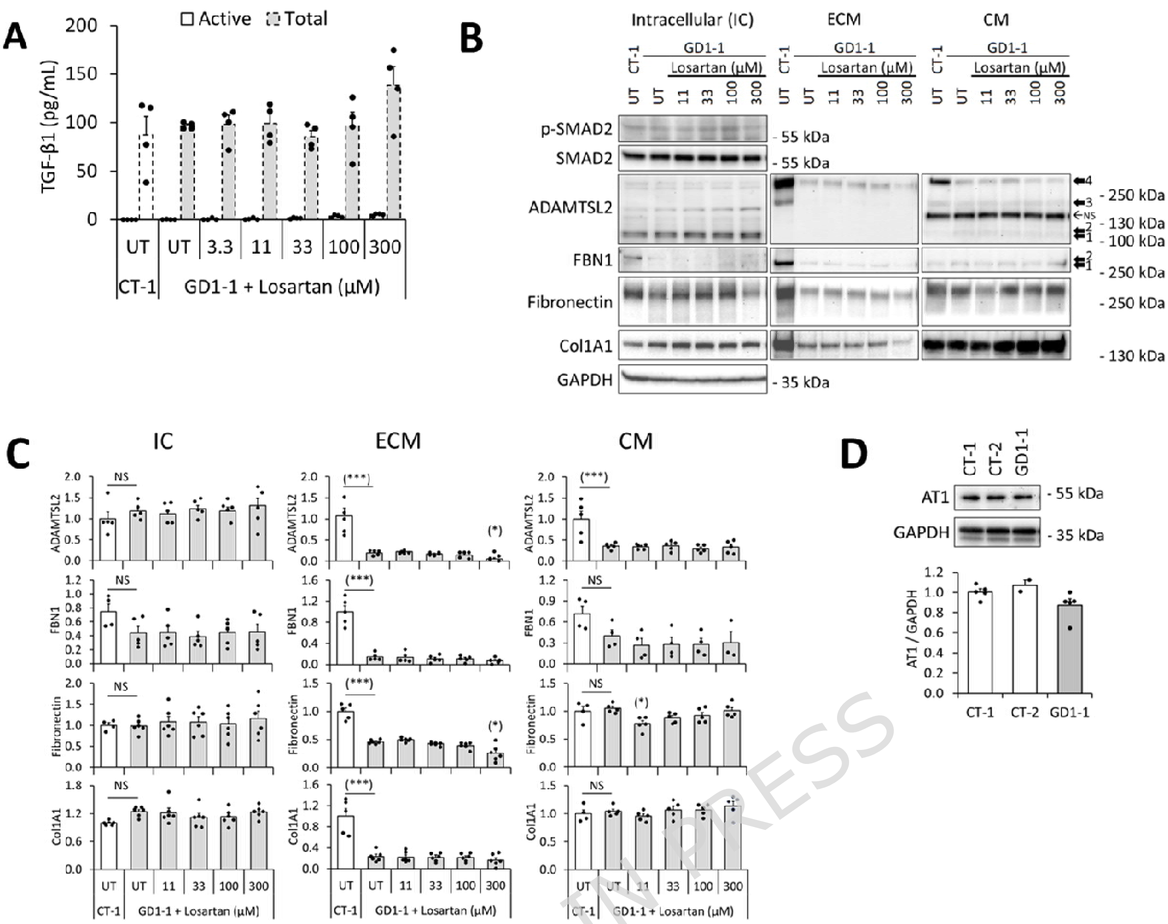

A
Active
Total
TGF-β1 (pg/mL)
UT
UT
3.3
11
33
100
300
CT-1
GD1-1 + Losartan (µM)
B
Intracellular (IC)
ECM
CM
CT-1
GD1-1
Losartan (µM)
p-SMAD2
SMAD2
ADAMTSL2
FBN1
Fibronectin
Col1A1
GAPDH
- 55 kDa
- 250 kDa
- 130 kDa
- 100 kDa
- 35 kDa
C
IC
ECM
CM
NS
(***)
(*)
D
CT-1
CT-2
GD1-1
AT1
GAPDH
AT1 / GAPDH

Table 1- Details of studied samples.

Sample ID	CT-1	CT-2	GD1-1	GD1-2	GD2-1	GD2-2	GD2-3
Diagnosis	CT	CT	GD1	GD1	GD2	GD2	GD2
Patient ID	GD016E	GD017E	GD001	GD006	GD004	GD009	GD012
Mutations			***ADAMTSL2*** c.182G>A (**p.R61H**) c.493G>A (**p.A165T**) compound heterozygous	***ADAMTSL2*** c.542T>C (**p.V181A**) c.707C>T (**p.P236L**) compound heterozygous	***FBN1*** c.5284G>A (**p.G1762S**) heterozygous	***FBN1*** c.5183C>T (**p.A1728V**) heterozygous	***FBN1*** c.5243G>C (**p.C1748S**) heterozygous
Sex	male	male	male	female	female	male	male
Age - years (months)	0 (1)	0 (6)	1 (11)	12 (0)	28 (9)	8 (8)	11 (0)

CT: Healthy control, GD1: Geleophysic Dysplasia type 1, GD2: Geleophysic Dysplasia type 2.